

MEMPHIS INTL, TN, USA

WMO: 723340

Lat: 35.056N

Lon: 89.987W

Elev: 77

StdP: 100.40

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13893

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.2	-4.9	-15.1	1.0	-4.4	-12.8	1.3	-2.8	11.0	8.8	9.6	10.5	4.2	10	0.464

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	35.8	25.0	34.6	24.7	33.6	24.5	26.7	33.0	26.1	32.2	25.6	31.4	3.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
25.0	20.3	29.9	24.4	19.6	29.5	23.9	18.9	29.1	83.7	32.9	81.2	32.3	78.9	31.5	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.3	7.5	DB	-10.3	37.4	2.6	1.7	-12.2	38.6	-13.7	39.6	-15.1	40.5	-17.0	41.8	(4)
(5)			WB	-11.5	27.7	2.4	0.8	-13.1	28.3	-14.5	28.7	-15.8	29.1	-17.6	29.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.5	5.6	7.8	12.3	17.5	22.3	26.6	28.0	27.9	24.5	18.2	11.7	7.2
	DBStd	9.25	6.26	5.96	5.58	4.50	3.72	2.60	2.19	2.41	3.52	4.70	5.05	5.55
	HDD10.0	479	167	103	39	4	0	0	0	0	0	2	41	124
	HDD18.3	1587	397	296	196	68	12	0	0	0	3	62	206	347
	CDD10.0	3223	29	42	112	228	382	498	558	554	436	256	91	36
	CDD18.3	1289	1	2	11	42	136	248	300	296	189	58	6	1
	CDH23.3	13363	1	3	59	293	1119	2680	3493	3418	1808	460	27	2
CDH26.7	5725	0	0	4	48	353	1187	1640	1629	734	128	2	0	
Wind	WSAvg	3.6	4.0	4.2	4.2	4.2	3.7	3.2	3.0	2.9	3.1	3.2	3.5	3.8
Precipitation	PrecAvg	1303	93	107	138	132	131	90	97	86	91	79	133	139
	PrecMax	1800	226	267	307	291	290	183	215	245	313	210	295	351
	PrecMin	873	14	28	38	41	21	13	31	16	16	1	19	30
	PrecStd	245	55	55	62	64	62	42	43	56	60	53	77	85
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.9	23.8	27.2	29.8	32.8	36.1	36.7	38.3	35.8	31.9	26.3	23.3
		MCWB	17.7	18.0	18.5	20.6	23.1	24.3	25.6	24.4	23.2	21.9	19.0	19.1
	2%	DB	19.8	21.6	25.0	28.1	31.4	34.6	35.4	36.1	33.9	29.7	23.6	20.3
		MCWB	16.6	16.7	17.3	19.6	22.4	24.3	25.4	25.2	23.6	21.2	17.5	16.8
	5%	DB	17.7	19.7	23.2	26.6	30.2	33.4	34.3	34.6	32.4	28.0	21.9	18.2
		MCWB	14.8	15.2	16.4	18.9	21.8	24.0	25.3	24.8	23.2	20.0	16.7	15.2
	10%	DB	15.5	17.5	21.1	25.0	28.8	32.2	33.2	33.2	30.8	26.0	19.9	16.1
		MCWB	12.2	13.4	15.2	18.2	21.4	23.7	25.0	24.5	22.3	19.1	15.6	13.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.7	19.6	20.1	22.7	25.1	26.8	27.4	27.5	25.7	24.2	20.7	19.8
		MCDB	21.1	22.0	24.6	27.4	30.2	32.5	33.7	34.2	31.9	29.4	23.5	22.2
	2%	WB	17.2	18.1	18.8	21.3	24.1	25.9	26.8	26.7	24.9	23.0	19.3	17.8
		MCDB	19.4	20.7	23.3	25.9	29.0	31.8	33.1	33.0	31.0	27.4	22.4	19.8
	5%	WB	15.5	16.3	17.6	20.4	23.2	25.3	26.3	26.0	24.2	21.8	17.9	15.8
		MCDB	17.5	19.0	21.7	24.8	28.1	31.3	32.5	32.0	29.8	25.8	20.9	17.8
	10%	WB	12.9	14.1	16.3	19.4	22.4	24.7	25.7	25.4	23.7	20.5	16.2	13.7
		MCDB	14.8	16.9	20.4	23.6	27.2	30.3	31.5	31.2	28.8	24.5	19.3	15.7
Mean Daily Temperature Range	5% DB	MDBR	8.9	9.3	10.1	10.3	9.7	9.5	9.1	9.5	10.2	10.6	9.9	8.6
		MCDBR	10.5	11.5	11.4	11.3	10.6	10.6	10.2	10.8	11.2	12.2	12.0	10.5
		MCWBR	8.3	8.1	6.7	5.6	4.5	3.8	3.5	3.4	3.9	5.6	7.6	8.1
	5% WB	MCDBR	9.5	10.1	9.9	9.3	9.4	9.3	9.4	9.4	9.8	10.0	9.2	
		MCWBR	8.3	8.2	6.9	5.6	4.6	4.2	3.8	3.7	3.9	5.4	8.0	8.0
Clear-Sky Solar Irradiance	taub	0.318	0.327	0.364	0.401	0.440	0.460	0.493	0.490	0.423	0.362	0.333	0.316	
	taud	2.496	2.467	2.392	2.284	2.234	2.187	2.130	2.121	2.304	2.451	2.483	2.517	
	Ebn at Noon	887	918	907	885	850	829	799	794	835	865	860	867	
	Edh at Noon	84	96	113	132	140	147	155	153	122	97	84	78	
All-Sky Solar Radiation	RadAvg	2.44	2.95	4.10	5.24	5.89	6.51	6.24	5.75	4.91	3.84	2.73	2.09	
	RadStd	0.20	0.40	0.37	0.34	0.35	0.39	0.32	0.37	0.36	0.46	0.28	0.21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (21 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page